InfiGUI-G1: Advancing GUI Grounding with Adaptive Exploration Policy Optimization

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Paper version: arXiv 2508.05731v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *inefficient exploration bottlenecks semantic alignment*

RLVR sharpens **spatial** alignment (precise pointing); **semantic** alignment — pointing at the *right* element — stalls on confident errors.

Why single-answer RLVR stalls

1. **Confidence trap.** RLVR resamples one wrong icon; the right one is never seen.
2. **One answer per pass.** No learning signal on hard samples.
3. **Diversity needs a brake.** Unguided multi-answer becomes collinear scans.

Evaluation setup

MMBench-GUI · ScreenSpot-Pro · UI-Vision · UI-I2E-Bench · ScreenSpot-V2 · Qwen2.5-VL 3B/7B · 44k samples · 5 runs, σ · all offline

- **All offline:** point-in-box top-1 accuracy.
- **Baseline:** Naive RLVR, same 44k data and hyperparameters.

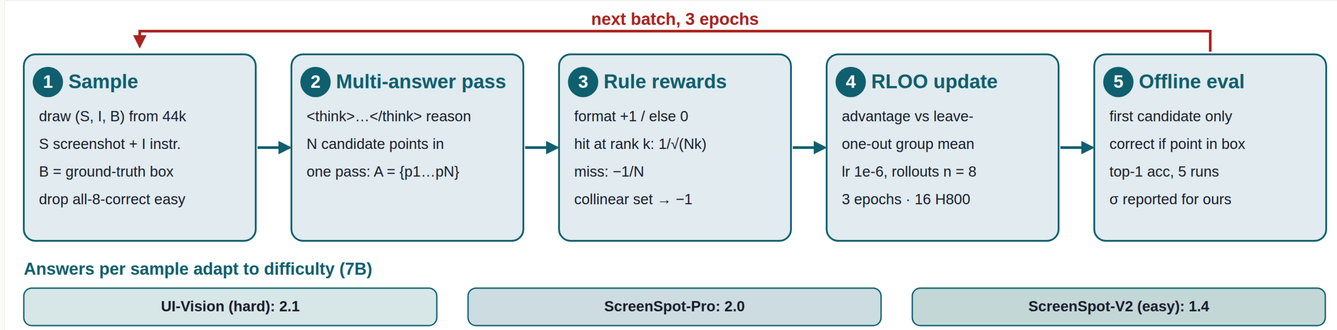
Our read (CUA Radar)

61	★★	offline
score of 100	band	evidence level

- **Reports:** offline eval; unseen apps; σ (5 runs); same-data baseline; ablation; code.
- **Flags:** no online eval; 3B/7B only; budget match partial.
- Pattern R01, rule-reward single-step RL. Not rerun.

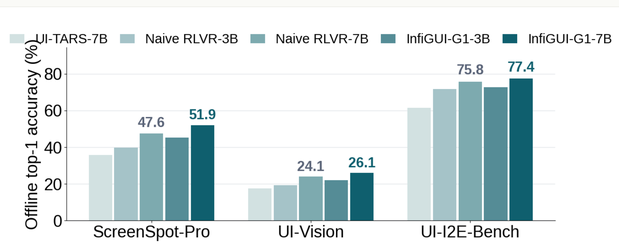
Method: AEPO — emit N points, reward by efficiency η = U/C

- **Sees / acts:** GUI screenshot + instruction in; one click point (x, y) out — single-step grounding, reasoning, Qwen2.5-VL backbones.
- **AEPO:** the model emits N candidate points in one forward pass; a deterministic rule reward scores the whole set; a +1 format reward gates it.
- **Signal:** reward 1/√(N·k) when the first point inside B ranks k-th; −1/N when all N miss; near-collinear set → −1; RLOO update.



Our redraw (not the paper's figure): the AEPO loop on 44k screenshots, and how N adapts at test time.

Key results: offline top-1 accuracy (%)



Tables 2–4, redrawn: offline top-1, first answer only; MMBench-GUI 7B 80.8, ScreenSpot-V2 7B 93.5.

Above same-data Naive RLVR on all five benchmarks, 3B and 7B.

SO WHAT → *MMBench-GUI 7B 80.8 and ScreenSpot-V2 7B 93.5, same protocol; 'w/ Expl. Success' rows are a different metric. σ ≤ 0.51 over 5 runs for ours.*

Ablation: ScreenSpot-Pro accuracy (%)

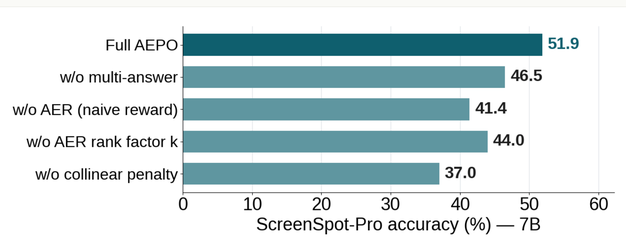


Table 6, redrawn: ScreenSpot-Pro, 7B, one component removed at a time; the 3B model shows the same order.

SO WHAT → *Each part matters. Removing the collinear penalty: accuracy 51.9 → 37.0 (7B) and answers per sample 2.0 → 8.2; 3B shows the same order (45.2 → 35.3, 2.1 → 6.6 answers). Without the rank factor k the model finds answers (expl. 47.6) but ranks them first too rarely (38.1). SFT rivals listed use >1M samples; this run uses 44k. The paper's reading: multi-answer enables exploration, AER shapes it, the rank factor k teaches confidence in what it finds, and the collinear penalty keeps exploration honest.*

Headline Numbers

51.9%
ScreenSpot-Pro 7B
vs 47.6 Naive RLVR-7B

- **80.8%** MMBench-GUI 7B
- **77.4%** UI-I2E-Bench 7B
- **93.5%** ScreenSpot-V2 7B

The paper's own controls

CONTROL	THIS PAPER	COMPARED WITH
One pass vs pass@4	58.0% at Avg. N = 2.0	Naive pass@4: 52.1%
Hard subset (7B)	17.4 (+61.1% rel.)	Naive RLVR 10.8
Data + budget	44k, RLOO n=8, 3 epochs	identical for Naive
5 runs, σ	ours: 0.05–0.51	baselines: 1 run

Unseen apps in the test sets

Takeaway

N candidates per pass + efficiency reward beat same-data Naive RLVR on five offline benchmarks, 3B and 7B.

44k samples · RLOO, 3 epochs · offline evidence only · code released

Scan to Read

Paper · Code